

PROJECT REPORT ON IOT-LINUX BASED ENVIRONMENTAL MONITORING

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CERTIFICATE

This is to certify that the project on **IOT-LINUX BASED ENVIRONMENTAL MONITORING SYSTEM** and term work carried out in the subject of Term Project is Bonafede work of **Dutt Panchal** (Roll no.: **EC028**) of B. Tech. semester V in the branch of **Electronics & Communication**, during the academic year 2024-25.

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ABSTRACT

This project presents a real-time **IoT-based Environmental Monitoring System** that measures temperature and humidity using the DHT11 sensor and transmits the data via Wi-Fi using the ESP8266 NodeMCU microcontroller. The collected data is stored in a local **MySQL database** using a Linux-based server (e.g., APACHE) and is displayed on a **web dashboard** using PHP and HTML. This system offers a cost-effective and scalable solution for environmental data monitoring with real-time updates and storage for future analysis.

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1. INTRODUCTION

1.1 Project Overview and Objective

The objective of this project is to design and implement a real-time environmental monitoring system that:

- Measures temperature and humidity from the surroundings.
- Transmits data wirelessly using ESP8266 to a server.
- Stores data in a database for logging and analysis.
- Displays the data on a web-based dashboard in real time.

This solution aims at creating a low-cost, scalable system for applications such as smart homes, agriculture, and weather monitoring.

1.2 Scope of the Project

- Real-time sensing and data acquisition.
- Wireless data transmission via Wi-Fi.
- Local data storage using MySQL.
- Dynamic web interface using PHP and HTML.
- Potential integration with cloud servers for remote access.
- Expansion to additional sensors and actuators for extended applications.

2. BLOCK DIAGRAM

Here is a block diagram for IoT-Linux Based Environmental Monitoring System:

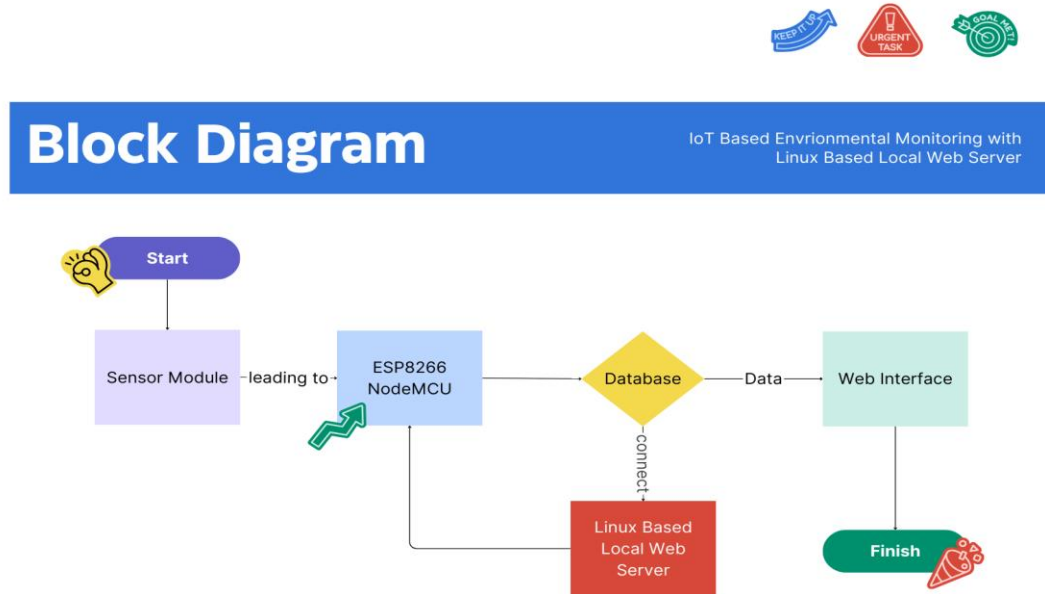


Figure 1 Block Diagram of IoT-Linux Based Environmental Monitoring System:

Explanation of Block Diagram: IoT-Based Environmental Monitoring with Linux-Based Local Web Server

This block diagram represents the **data flow and working** of the environmental monitoring system you built using ESP8266, DHT11, and a Linux local server. Here's a **step-by-step explanation** of each block:

1. Start

- The process begins with the **Sensor Module** being powered on and ready to collect environmental data (temperature and humidity).

2. Sensor Module

- This module consists of the **DHT11 sensor**, which detects real-time **temperature and humidity** from the surrounding environment.
 - It acts as the primary data source in the system.
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3. ESP8266 NodeMCU

- The NodeMCU receives the data from the DHT11 sensor.
 - It **processes the data** and **connects to Wi-Fi** to communicate with the Linux-based server.
 - The NodeMCU sends this data to the **MySQL database** on the server via HTTP POST requests (using PHP script `insert_data.php`).
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4. Linux-Based Local Web Server

- This is the **host server** (e.g., a PC running Apache, MySQL, PHP stack - XAMPP or LAMP).
 - It processes incoming data via PHP scripts and **stores it in the MySQL database**.
 - The server also hosts the **web interface** (e.g., `index.php`) to display the stored data.
 - The ESP8266 sends data directly to this server's IP over the local network.
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5. Database

- The **MySQL database** is used to **store all incoming temperature and humidity values** with timestamps.
 - It acts as a data logger for analysis and visualization.
-

6. Web Interface

- A **PHP and HTML-based dashboard** that fetches data from the database and displays it in real-time.
 - Users can **access the interface on any device** connected to the same network by entering the server's IP address in the browser.
-

7. Finish

- The system completes one cycle of sensing, storing, and displaying data.
 - The process continues in a **loop** with the ESP8266 sending updated data periodically (e.g., every 5 seconds).
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Flow Summary:

Sensor → NodeMCU → Local Server (Database) → Web Interface

- The diagram visually outlines how each part of the system is **interconnected**, from sensing to data display.
- It emphasizes **local server deployment**, making it suitable for offline/local environments without relying on cloud services.

3. COMPONENT DESCRIPTION

Here are the list of components (with brief description and specifications) that we used to build this project:

DHT11 Sensor (Temperature and Humidity Sensor)

- A low-cost digital sensor used to measure **temperature and humidity**.
- It provides **digital output** directly to microcontrollers.
- Operates on **3.3V to 5V** supply and is ideal for environmental monitoring projects.

ESP8266 NodeMCU (Wi-Fi Module)

- A low-power **Wi-Fi microcontroller** module used to connect sensors to a network.
- It reads data from the DHT11 sensor and **sends it via HTTP POST** to a local server.
- Supports **programming via Arduino IDE** and includes GPIO pins for sensor interfacing.

Linux-Based Local Server (Laptop/PC with XAMPP or LAMP)

- A system running **Apache (web server), MySQL (database), and PHP** to handle HTTP requests and data storage.
- Hosts the **web interface** and stores data received from the ESP8266.
- **Wi-Fi Network (Router/Mobile Hotspot)**
- Provides **network connectivity** for ESP8266 and server.
- Ensures smooth **data communication over HTTP** in the local network.

4. CIRCUIT DIAGRAM

Here is a circuit diagram for IoT Based Smart Environmental Monitoring:

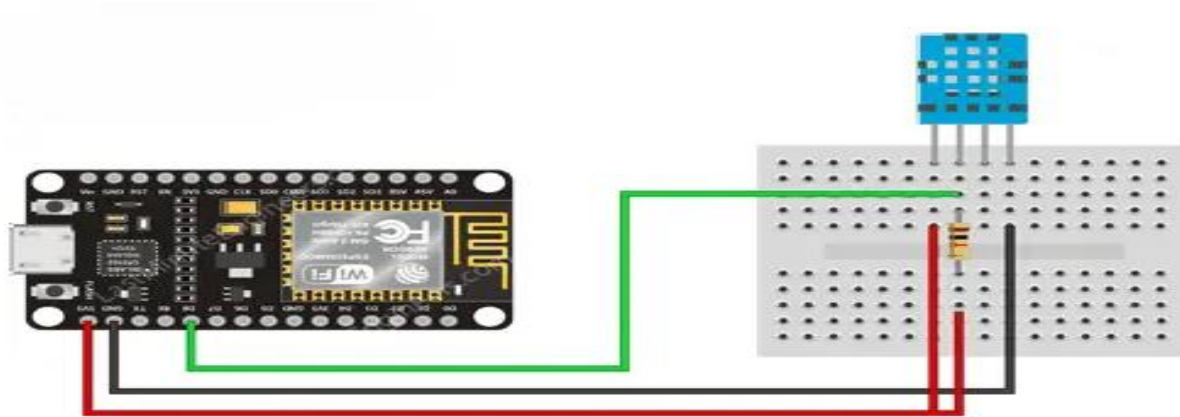


Figure 2 Circuit Diagram

This circuit diagram shows the interfacing of a DHT11 Temperature and Humidity Sensor with an ESP8266 NodeMCU board using a breadboard for prototyping.

Connections:

DHT11 Sensor:

- The DHT11 has **three pins** used here:
- **VCC (leftmost pin)** → Connected to **3.3V** on the NodeMCU (Red wire).
- **DATA (middle pin)** → Connected to **GPIO D4** (which is internally GPIO2) on the NodeMCU (Green wire).
- **GND (rightmost pin)** → Connected to **GND** on the NodeMCU (Black wire).

Resistor:

- A **10kΩ pull-up resistor** is connected between the **DATA pin and VCC**.
- Purpose: Ensures **stable data communication** between the DHT11 and ESP8266 by pulling the data line high when idle.

ESP8266 NodeMCU:

- Acts as the **Wi-Fi-enabled microcontroller**, which reads the sensor data.
- Sends temperature and humidity data via **HTTP POST** to the **local web server**.

Breadboard:

- Used to organize the circuit in a **non-permanent** manner.
- Allows easy troubleshooting and changes to the wiring during testing.

Power Supply:

- NodeMCU is powered via **USB connection** to a laptop or power adapter.
- The USB provides **5V input**, but internally, the NodeMCU regulates it to **3.3V** required by DHT11.

Working Summary:

- The sensor continuously measures **real-time temperature and humidity**.
- The NodeMCU reads this data from the **DATA pin**.
- It connects to the **Wi-Fi network**, then sends the data to the **Linux-based local server** for **storage and web visualization**.

5. PRACTICAL SETUP AND TROUBLESHOOTING

Setup Steps:

1. Connect DHT11 to NodeMCU.
2. Upload the Arduino code to NodeMCU using the Arduino IDE.
3. Configure Wi-Fi credentials and server IP in code.
4. Run Apache and MySQL on the Linux server.
5. Place `insert_data.php` and `index.php` in the server's root directory.
6. Access the dashboard via `http://<server_ip>/index.php`.

Common Troubleshooting:

- HTTP Response Code -1: Check Wi-Fi connection and server IP.
- Data not updating: Check database insertion and dashboard refresh interval.
- No sensor reading: Verify sensor wiring and use `Serial Monitor` for debugging.

6. Result and Analysis

Real-time temperature and humidity data are successfully collected every 5 seconds.

- Data is stored in the MySQL database and displayed in a user-friendly web interface.
- System tested on different Wi-Fi networks and maintained reliable performance.

7. LIMITATIONS OF THE CIRCUIT

Limited range due to reliance on Wi-Fi connectivity.

- Accuracy limited by DHT11 sensor ($\pm 2^{\circ}\text{C}$, $\pm 5\%$ RH).
- Requires constant power supply for server and NodeMCU.
- Localhost access only unless configured for external access or cloud integration.

8. APPLICATION AND FUTURE WORK

Applications:

- Smart home environmental monitoring.
- Agricultural field monitoring.
- Industrial temperature/humidity control.
- School/office indoor air quality monitoring.

Future Work:

- Use of advanced sensors (e.g., DHT22, BME280) for higher accuracy.
 - Integration with cloud platforms (e.g., AWS, Firebase).
 - Add alert systems via SMS or email.
 - Mobile app interface for remote access.
 - Solar-powered system for outdoor deployment.
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9. CONCLUSION

This project successfully demonstrates a functional **IoT-based Environmental Monitoring System** that captures real-time temperature and humidity data, stores it locally, and displays it via a web interface. The system offers scalability and can be extended for more sensors and applications in future developments.

10. REFERENCES

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